

PROGRAM 24

AIM

To implement the Monkey and Banana Problem using Prolog to find a sequence of actions by which a monkey can reach and obtain a banana.

ALGORITHM

1. Start the program.
2. Define the initial position of the monkey, box, and banana.
3. Define the possible actions such as moving, pushing the box, climbing the box, and grabbing the banana.
4. Check whether the monkey is at the same position as the box.
5. If not, move the monkey towards the box.
6. Push the box to the position below the banana.
7. Allow the monkey to climb onto the box.
8. Check whether the monkey is below the banana.
9. If yes, allow the monkey to grab the banana.
10. Display the sequence of actions.
11. Stop the program.

PROGRAM

```
move(monkey, door).
move(monkey, window).
push(box, door, window).
climb(monkey, box).
get_banana(monkey, banana).
solve :-
    move(monkey, door),
    push(box, door, window),
    climb(monkey, box),
    get_banana(monkey, banana),
    write('Monkey gets the banana.').
```

OUTPUT

```
39 ?- consult(['c:/Users/Hema Neepe/OneDrive/Documents/Monkey_Banana.pl']).  
true.
```

```
40 ?- solve.  
Monkey gets the banana.  
true.
```

RESULT

The Monkey and Banana Problem was successfully implemented in Prolog, and the monkey was able to reach and obtain the banana through the defined sequence of actions.